Wenjuan Lin

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Education

University of Illinois Urbana-Champaign *Sep. 2023 – Jun. 2027*
Bachelor of Science in Computer Engineering

- GPA: 3.98/4.00

Zhejiang University *Sep. 2023 – Jun. 2027*
Bachelor of Engineering in Electrical and Computer Engineering

- GPA: 3.94/4.00, Rank: 2/71

Honors and Awards

National Scholarship *2024*
Awarded by the Ministry of Education of China for outstanding academic and overall performance.

Dean's List *2024, 2025*
Awarded for outstanding academic performance at the University of Illinois Urbana-Champaign.

Second Prize Scholarship for Academic Excellence *2024, 2025*
Awarded by ZJUI; granted to only two students in the major annually.

Second Prize & Best Essay Prize in Concrete Dragon Boat Race *2025*
Awarded by Zhejiang University for outstanding performance in the Concrete Dragon Boat Race competition.

Meritorious Winner in Mathematical Contest in Modeling *2024*
Awarded to teams ranking in the top 9% worldwide for outstanding performance in mathematical modeling.

Research Experience

Underwater Navigation under Communication and Visibility Constraints *Zhejiang, China*
Advisors: Wenguan Wang, Zhejiang University *Jun. 2026 – Present*

- Conducting research on object-goal navigation for autonomous underwater agents in communication- and visibility-constrained environments.
- Exploring the integration of pretrained foundation models to improve perception, reasoning, and navigation performance.
- Studying the adaptation of indoor VLN and ObjectNav methodologies to underwater domains.

Vision-Language-Action Model Exploration *Zhejiang, China*
Advisors: Siliang Tang, Guoming Wang, Zhejiang University *Jun. 2025 – Aug. 2025*

- Conducted systematic study of recent Vision-Language-Action (VLA) and world models papers.
- Reproduced open-source VLA pipelines under constrained computational resources, building hands-on experience in experiment reproduction, debugging, model training, and performance evaluation.
- Awarded **Outstanding Project** by ZJU–UIUC.

Course Projects

Autonomous Drone Racing | 1st Place in the Drone Racing Competition *Illinois, United States*
ECE484 Principles of Safe Autonomy, Instructor: Sayan Mitra *Project Website*

- Developed a ROS 2-based autonomous drone racing pipeline for Crazyflie 2.1, integrating IMU and motion capture (MoCap) sensors.
- Designed a classical controller (sparse planning + PID control), with systematic parameter tuning, achieving the fastest hardware racing time among all teams.
- Trained a PPO policy using JAX in a drone racing simulator, outperforming the classical controller in simulation and successfully deploying it on simple real-world tracks.

GPT-2 CUDA Kernel Optimizations | Top 3 in Performance Competition *Illinois, United States*
ECE408 Applied Parallel Programming, Instructor: Volodymyr Kindratenko *Project Report*

- Implemented CUDA kernel optimizations including Tensor Cores, FlashAttention, and warp-level reduction

techniques for GPT-2 acceleration.
  - Identified compute, memory, and warp-level bottlenecks using NVIDIA Nsight Systems and Nsight Compute.

Land.io FPGA Two-Board Game | Top 3 Final Project — *Illinois, United States*
ECE385 Digital Systems Laboratory, Instructor: Zuofu Cheng — *Project Report*
  - Built a real-time two-player FPGA game across synchronized boards using SystemVerilog and MicroBlaze.
  - Implemented a master–slave SPI protocol for inter-board synchronization, along with custom AXI4-Lite peripherals, HDMI rendering, and DDR3-backed video/audio playback.

Extracurricular Activities

EKF-Based State Estimation for a Student Rocket — *Illinois, United States*
Guidance, Navigation & Control Team Member | Illinois Space Society — *Sep. 2025 – Feb. 2026*
  - Contributed to debugging and validation of a student rocket state-estimation pipeline based on Extended Kalman Filtering, identifying modeling and coordinate-frame inconsistencies.
  - Participated in the LUNA 1.0 launch and served as a Launch Operations team member for LUNA 2.0.

Competitions

Second Prize in Sixth College Student Concrete Dragon Boat Race — *Zhejiang, China*
Bo Yan Team, Advisor: Shilei Zhang — *Project Poster*
  - Designed and implemented the electrical control system and documented the system design.
  - Collaborated with four teammates to design, build, and test a concrete dragon boat.
  - Competed in both sprint and obstacle racing events.

Meritorious Winner in Mathematical Contest in Modeling — *Zhejiang, China*
Teammates: Siyuan Gong, Xinyi Zheng, Advisor: Wee-Liat Ong — *Project Report*
  - Developed the Sea Lamprey Population Model using Fisher's Principle, the Leslie Matrix, and the Lotka-Volterra Model to simulate the dynamics of sea lamprey populations and their impact on ecosystems.
  - Utilized MATLAB to stimulate difference equations and visualize the results of the mathematical model through various plots.
  - Wrote the report presenting the final results, addressing how sex ratio variation influences both sea lamprey populations and the surrounding ecosystem.

Work

Main Manager in Publicity Department — *Zhejiang, China*
ZJU-UIUC Institute — *Jul. 2024 – Jun. 2025*
  - Managed the student publicity team, overseeing recruitment, interviews, and regular meetings with faculty.
  - Led the planning and production of high-quality visual content, including WeChat articles, videos, and promotional graphics.

Writing Assistant in Rhetoric 102 — *Zhejiang, China*
Instructor: Mary Lucille Hays — *Feb. 2025 – Jun. 2025*
  - Assisted students in developing academic writing, argumentation, and revision strategies.

Skills

Programming Languages: C++, Python, C, MATLAB, CUDA, Verilog
Frameworks & Tools: PyTorch, ROS 2, Linux, Git
Languages: Chinese (Native), English (TOEFL 5.0/6.0), Cantonese (Basic)